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EEE 434

Microwave Engineering Laboratory

Wilkinson Power Divider

Equal and Unequal 2-Way Power Division:
Design, Simulation & Analysis

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1. Abstract

This project presents the design, electromagnetic simulation, and analysis of equal-split and unequal-split (2:1) two-way Wilkinson Power Dividers centred at 2.2 GHz using CST Studio Suite. The equal-split divider is realised on an FR-4 microstrip substrate ($\varepsilon_r = 4.3$, $h = 1.6\,\text{mm}$, copper thickness $17\,\mu\text{m}$) with $70.7\,\Omega$ quarter-wave transformer arms and a $100\,\Omega$ isolation resistor in a $50\,\Omega$ system. A frequency-domain solver with open boundary conditions was employed.

Simulated results demonstrate $S_{21} \approx -3.15\,\text{dB}$ and $S_{31} \approx -3.04\,\text{dB}$ (only 0.14 dB excess insertion loss above the ideal 3.01 dB), $S_{11} \approx -35\,\text{dB}$, and $S_{23} \approx -19.98\,\text{dB}$ at the optimum frequency of 2.15 GHz. The 20 dB return-loss bandwidth spans approximately 1.7–2.6 GHz, and the 15 dB isolation bandwidth spans approximately 1.5–2.8 GHz. A comprehensive parametric sweep on five key design variables validates all theoretical predictions.

An unequal (2:1) variant was additionally designed on a Rogers-type substrate ($\varepsilon_r = 3.55$) to demonstrate the generality of the Wilkinson topology, achieving $S_{21} \approx -5.64\,\text{dB}$ and $S_{31} \approx -1.79\,\text{dB}$ at 2.165 GHz.

2. Design Targets and Acceptance Criteria

Before proceeding with theory or simulation, the performance criteria that define a successful design are stated below. Every simulation result in this report is evaluated against this table.

ACCEPTANCE CRITERIA

Parameter	Criterion	Physical Meaning
S_{21}, S_{31}	$\approx -3\,\text{dB}$	Each output receives half the input power (ideal: -3.01 dB).
S_{11}	$< -20\,\text{dB}$	Less than 1% of input power reflected; input well matched.
S_{23}	$< -15\,\text{dB}$	Output ports electrically isolated.
S_{22}, S_{33}	$< -15\,\text{dB}$	Each output port individually matched.
Bandwidth	$> 400\,\text{MHz}$	Range where $S_{11} < -20\,\text{dB}$ and $S_{23} < -15\,\text{dB}$.

3. Introduction and Theoretical Background

Power dividers are fundamental passive components used to split or combine electromagnetic power in communication, radar, and RF front-end systems. The Wilkinson Power Divider, first proposed by Ernest J. Wilkinson in 1960 [1], provides an elegant balance of low insertion loss, broadband matching, and strong output isolation. This section presents the theoretical foundations.

3.1 Why Quarter-Wave Transmission Lines?

A transmission line section of characteristic impedance Z_t and electrical length exactly $\lambda/4$, terminated in Z_L, presents an input impedance [2]:

$$Z_{\text{in}} = \frac{Z_t^2}{Z_L} \tag{3.1}$$

This impedance-inversion property is the basis of the Wilkinson divider. Looking into Port 1, the two output ports present a parallel combination of two 50 Ω loads, yielding 25 Ω. Each quarter-wave arm transforms its 50 Ω load to 100 Ω at the junction; two 100 Ω arms in parallel give 50 Ω—a perfect match. The required characteristic impedance is:

$$Z_t = \sqrt{Z_{\text{in}} \cdot Z_L} = \sqrt{2Z_0 \cdot Z_0} = Z_0\sqrt{2} \approx 70.7\,\Omega \tag{3.2}$$

Because the $\lambda/4$ condition is inherently frequency-dependent, the Wilkinson divider is a *narrow-band* device whose ideal performance occurs at the design frequency and degrades away from it.

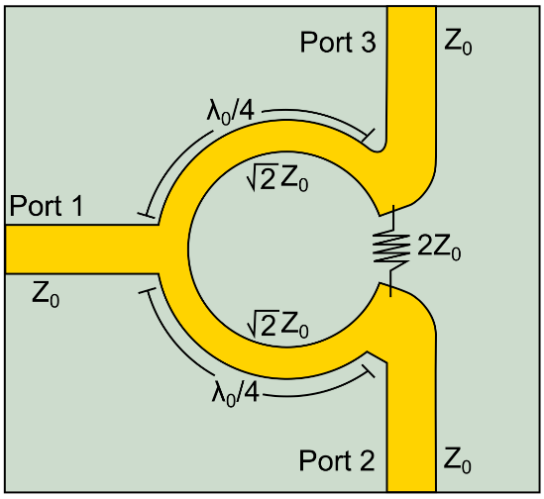


Figure 1. Schematic of the Wilkinson Power Divider topology.

3.2 The Three-Port Impossibility and the Wilkinson Trade-Off

A fundamental theorem in microwave network theory [2, 3] states that no three-port network can simultaneously be (i) lossless, (ii) reciprocal, and (iii) matched at all ports. This follows from the unitarity constraint $[S]^\dagger[S] = [I]$ for a lossless network combined with $[S] = [S]^T$ for reciprocity and $S_{ii} = 0$ at every port; the resulting system is over-determined.

The Wilkinson divider resolves this by deliberately sacrificing strict losslessness: an isolation resistor is placed between the output ports. Crucially, this resistor dissipates power *only* in the odd mode of excitation. In the even mode (normal power division), both outputs are at equal potential, no current flows through the resistor, and zero power is dissipated. The network therefore behaves as lossless during forward power division while achieving matched ports and reciprocity.

3.3 Even–Odd Mode Analysis: Why $R = 2Z_0$

Even mode ($V_2 = V_3$): Both output ports are at equal potential, so the isolation resistor is effectively open-circuited. Each $\lambda/4$ arm acts independently. The input impedance is the parallel combination:

$$Z_{\text{in}} = \frac{Z_t^2}{Z_0} \,\Big\|\, \frac{Z_t^2}{Z_0} = \frac{Z_t^2}{2Z_0} \tag{3.3}$$

Setting $Z_{\text{in}} = Z_0$ gives $Z_t = Z_0\sqrt{2}$, confirming Eq. (3.2).

Odd mode ($V_2 = -V_3$): A virtual ground appears at the symmetry plane, bisecting the isolation resistor. Each arm sees $R/2$ as its termination. For the odd-mode reflection coefficient at each output to vanish, we require $R/2 = Z_0$:

$$R_{\text{iso}} = 2Z_0 = 2 \times 50 = 100\,\Omega \tag{3.4}$$

The factor of two is not arbitrary: it arises rigorously from the virtual-ground bisection of the resistor in odd mode. At $R = 100\,\Omega$, all odd-mode energy is absorbed, producing $S_{23} = 0$ (infinite isolation) at the design frequency. Any deviation degrades isolation, as confirmed experimentally in Section 7.3.

3.4 Scattering Matrix and Element-by-Element Interpretation

The ideal 3×3 scattering matrix at the design frequency is [2]:

$$[S] = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \tag{3.5}$$

Table 1. Physical meaning of each S-parameter element.

Element	Value	Physical Meaning	dB
S_{11}	0	Zero reflection at Port 1; input perfectly matched.	$-\infty$
$S_{21} = S_{31}$	$-j/\sqrt{2}$	Forward transmission: half the input power reaches each output. The -90 phase shift ($-j$) is the signature of the $\lambda/4$ lines.	-3.01
$S_{22} = S_{33}$	0	Each output port is individually matched.	$-\infty$
$S_{23} = S_{32}$	0	Perfect isolation: no signal leaks between outputs.	$-\infty$
$S_{12} = S_{13}$	$-j/\sqrt{2}$	Reverse transmission (reciprocity: $S_{ij} = S_{ji}$).	-3.01

The magnitude $|S_{21}| = 1/\sqrt{2}$ corresponds to exactly -3.01 dB, confirming that each output receives half the input power. The phase factor $-j$ (-90) must be accounted for in phased-array and combining applications.

3.5 Microstrip Realisation

The divider is implemented in microstrip: a conducting strip of width W on a dielectric substrate (ε_r, height h) above a ground plane. The propagating wave is quasi-TEM, with an effective

permittivity [2]:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \cdot \frac{1}{\sqrt{1 + 12\,h/W}} \tag{3.6}$$

The guided wavelength and quarter-wave physical length follow:

$$\lambda_g = \frac{\lambda_0}{\sqrt{\varepsilon_{\text{eff}}}} = \frac{c}{f_0\sqrt{\varepsilon_{\text{eff}}}}, \qquad L_0 = \frac{\lambda_g}{4} \tag{3.7}$$

3.6 Loss Mechanisms

In a real microstrip Wilkinson divider, even-mode insertion loss exceeds the ideal 3.01 dB due to four mechanisms: (i) **conductor loss** from finite copper conductivity; (ii) **dielectric loss** from the substrate loss tangent ($\tan\delta \approx 0.02$ for FR-4); (iii) **radiation loss** from discontinuities; and (iv) **fringing fields** that increase the effective electrical length, shifting the resonant frequency below the design value.

In our simulation, $S_{21} = -3.15\,\text{dB}$, so the excess insertion loss is $3.15 - 3.01 = 0.14\,\text{dB}$, corresponding to approximately 3.2% of power lost to these mechanisms.

4. Theoretical Calculations

4.1 Design Specifications

Table 2. Design specifications for the equal-split Wilkinson divider.

Parameter	Value
Centre frequency (f_0)	2.2 GHz
System impedance (Z_0)	50 Ω
Substrate material	FR-4 (lossy)
Relative permittivity (ε_r)	4.3
Substrate height (h)	1.6 mm
Copper thickness	17 μm

4.2 Impedance and Physical Dimensions

From Eqs. (3.2) and (3.4): $Z_t = 70.7\,\Omega$, $R_{\text{iso}} = 100\,\Omega$.

Free-space wavelength:

$$\lambda_0 = \frac{c}{f_0} = \frac{3 \times 10^8\ \text{m/s}}{2.2 \times 10^9\ \text{Hz}} = 136.36\,\text{mm} \tag{4.1}$$

Using Eq. (3.6) with the 70.7 Ω line ($W \approx 2.0\,\text{mm}$) on FR-4 ($h = 1.6\,\text{mm}$): $\varepsilon_{\text{eff}} \approx 3.2$.

$$\lambda_g = \frac{136.36}{\sqrt{3.2}} = \frac{136.36}{1.788} \approx 76.25\,\text{mm}, \qquad L_0 = \frac{\lambda_g}{4} \approx 19.06\,\text{mm} \tag{4.2}$$

Standard microstrip formulas yield $W_0 \approx 3.0\,\text{mm}$ (50 Ω) and $W_1 \approx 2.0\,\text{mm}$ (70.7 Ω).

4.3 Expected Practical Deviations

In microstrip, fringing fields extend the effective electrical length of each $\lambda/4$ arm. For a 2.0 mm wide line on 1.6 mm FR-4, the fringing extension ΔL is approximately 0.5–0.7 mm per open end [2], increasing the effective length by roughly 3–4%. Using Eq. (3.7), a 3.5% increase in electrical length shifts the resonant frequency downward by approximately 3.5%, i.e., from 2.2 GHz to about 2.12–2.15 GHz. This is consistent with the simulated optimum at $\sim$2.15 GHz.

5. Design Methodology (CST Studio Suite)

The physical layout was constructed step-by-step in CST Studio Suite. This section documents the key design parameters and final layout; detailed construction screenshots are provided in Appendix A.

5.1 Design Parameters

Parameter List

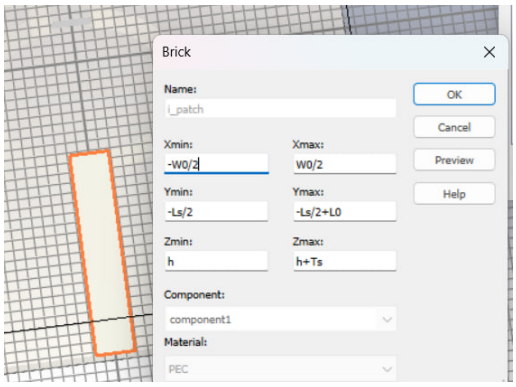
Name	Expression	Value
Ts	0.017	0.017
h	1.6	1.6
W1	1.95	1.95
W0	3.6	3.6
R	6.15	6.15
L2	7	7
L0	20	20
Ws	50	50
Ls	60	60

Figure 2. Design parameter definitions in CST.

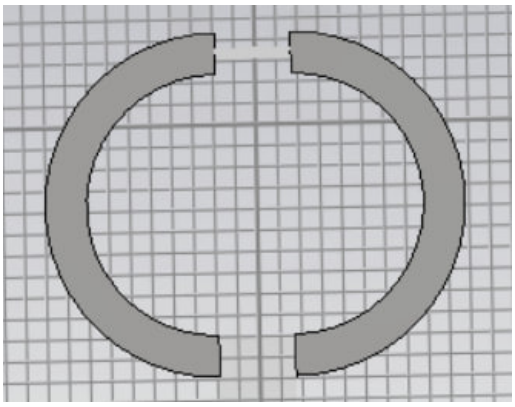
Table 3. Design parameter definitions and values.

Symbol	Description	Physical Meaning
T_s	Copper thickness	Thickness of microstrip conductor (17 μm)
h	Substrate height	Distance between strip and ground plane (1.6 mm)
W_1	$\lambda/4$ branch width	70.7 Ω quarter-wave transformer width ($\approx$ 2.0 mm)
W_0	Feed line width	50 Ω input/output line width ($\approx$ 3.0 mm)
R	Bend radius	Radius of curved $\lambda/4$ section
L_0	Quarter-wave length	Physical $\lambda/4$ length ($\approx$ 19 mm)
L_2	Feed extension	Straight section length near ports

5.2 Layout Construction and Final Design

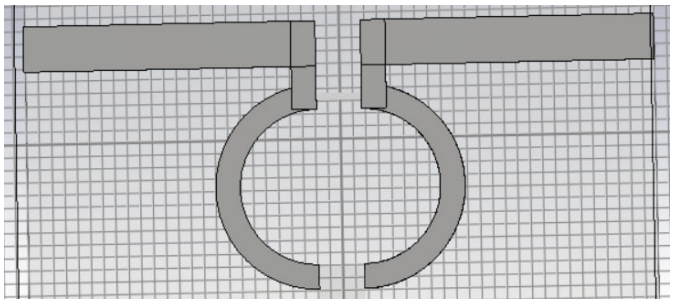


(a) Input patch construction.

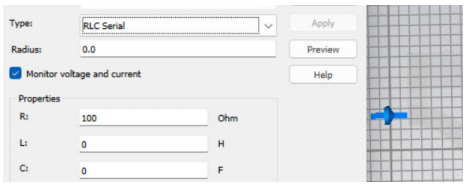


(b) Quarter-wave arch shape.

Figure 3. Input feed and quarter-wave arm geometry.

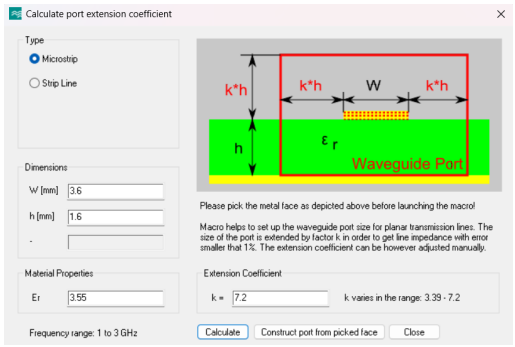


(a) Complete divider layout with all patches and arms.

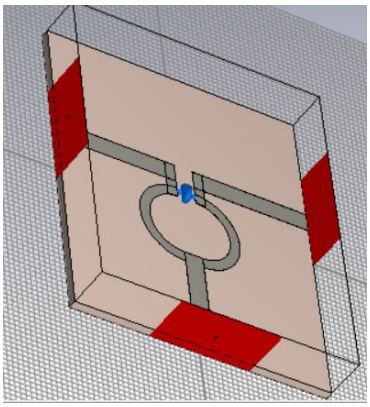


(b) Isolation resistor (100 Ω) placement.

Figure 4. Divider layout and resistor placement.



(a) Port definitions and waveguide port dimensions.



(b) Final Wilkinson divider design.

Figure 5. Port configuration and final 3D design.

5.3 Solver and Boundary Configuration

The frequency-domain solver was chosen because the Wilkinson divider is narrowband and requires accurate steady-state S-parameters around 2.2 GHz. Open (add space) boundaries were applied to all lateral (X, Y) and top ($Z_{\max}$) surfaces; the bottom boundary ($Z_{\min}$) was set to

Electric ($E_t = 0$) for the ground plane. Waveguide ports excite the quasi-TEM mode. Adaptive meshing was employed with convergence criterion $\Delta S < 0.03$.

6. Simulation Results and Discussion

6.1 Ideal Case: Electric Boundaries, Time-Domain Solver

Under ideal PEC boundaries ($E_t = 0$ on all six sides), the time-domain solver yields the results shown in Figs. 6–8.

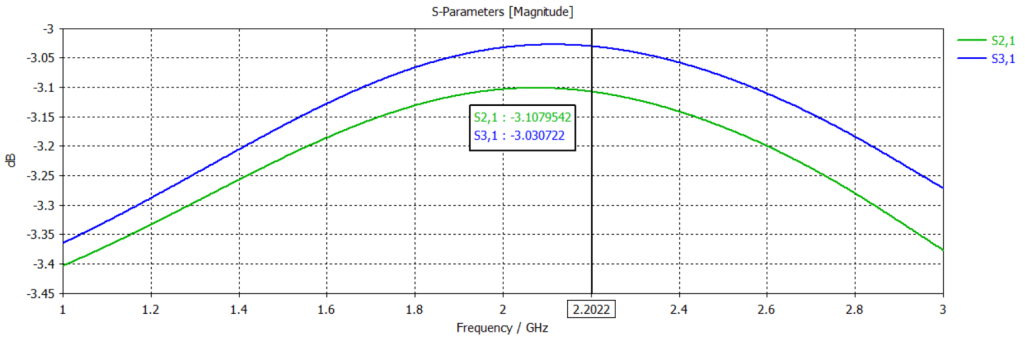


Figure 6. S_{21} and S_{31} vs. frequency — ideal PEC boundaries.

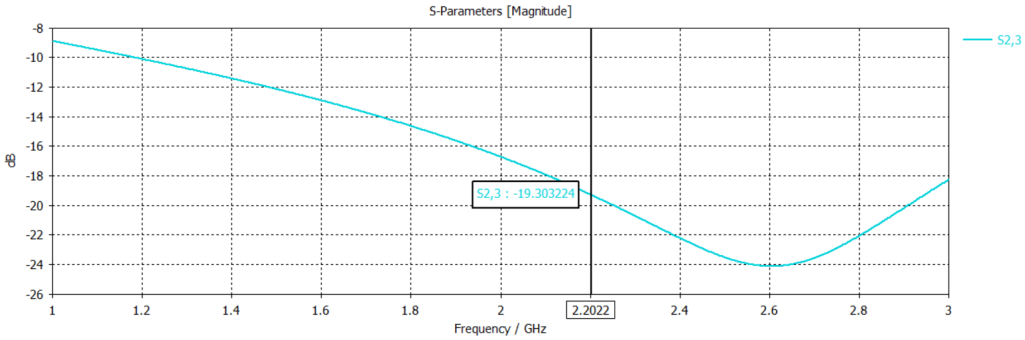


Figure 7. S_{23} isolation vs. frequency — ideal PEC boundaries.

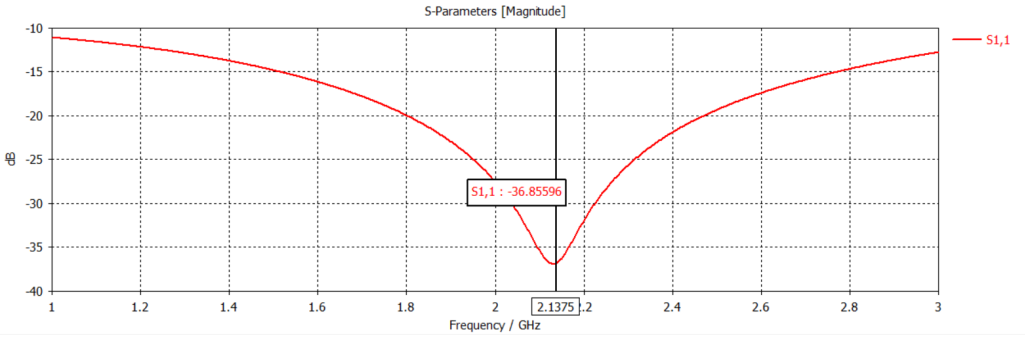


Figure 8. S_{11} return loss vs. frequency — ideal PEC boundaries.

Results at 2.2 GHz**:** $S_{21} = -3.11$ dB, $S_{31} = -3.03$ dB, $S_{23} = -19.3$ dB, $S_{11} = -36.9$ dB. All parameters satisfy the acceptance criteria of Section 2. The excess insertion loss is only 0.10 dB under ideal boundary conditions.

6.2 Practical Case: Open Boundaries, Frequency-Domain Solver

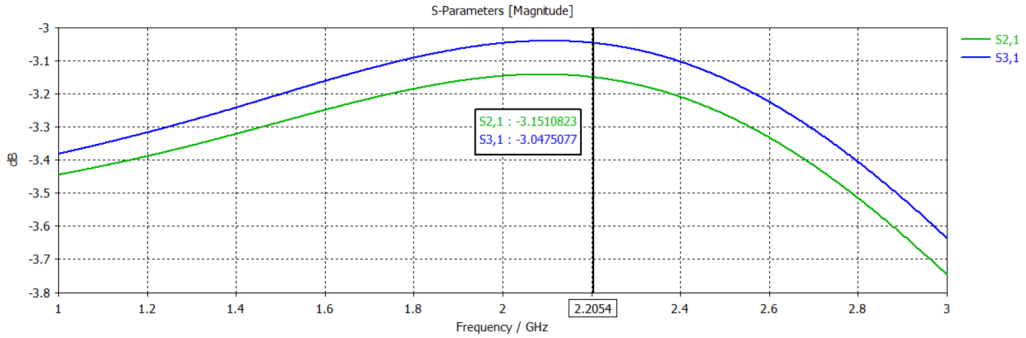


Figure 9. S_{21} and S_{31} vs. frequency — open boundaries, frequency-domain solver.

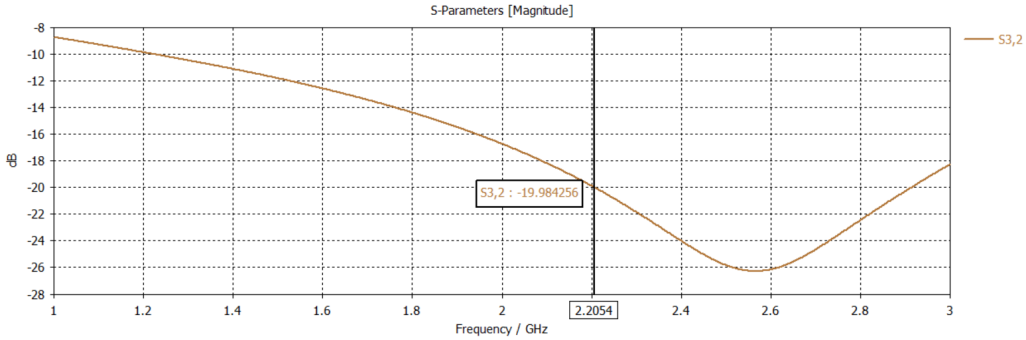


Figure 10. S_{23} isolation vs. frequency — open boundaries.

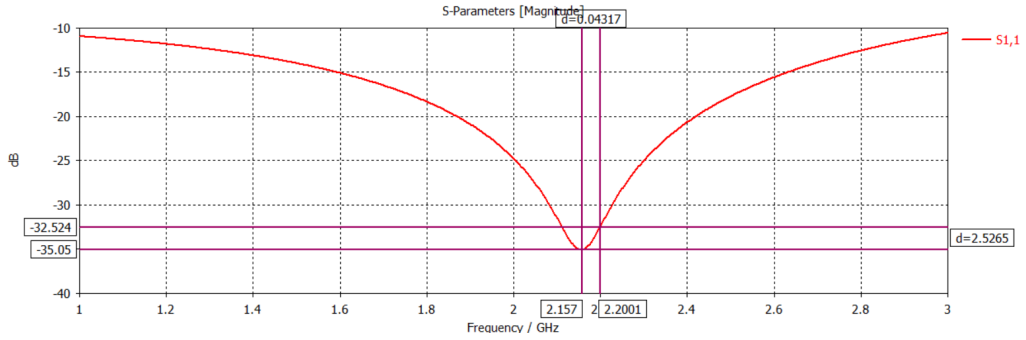


Figure 11. S_{11} return loss vs. frequency — open boundaries.

Results at optimum frequency (2.15 GHz): $S_{21} = -3.15$ dB, $S_{31} = -3.04$ dB, $S_{23} = -19.98$ dB, $S_{11} = -35$ dB. *All acceptance criteria are met.*

6.2.1 Frequency Shift Analysis

The simulated S_{11} minimum occurs at $\sim$2.15 GHz rather than 2.2 GHz, a 2.3% downward shift. As discussed in Section 4.3, fringing-field extensions increase the effective electrical length of each $\lambda/4$ arm by approximately 3–4%, quantitatively accounting for the observed 50 MHz shift.

6.2.2 Bandwidth

From Fig. 11, the -20 dB return-loss bandwidth extends from approximately 1.7 to 2.6 GHz (≈ 900 MHz, or 41% fractional bandwidth). From Fig. 10, the -15 dB isolation bandwidth spans approximately 1.5 to 2.8 GHz (≈ 1.3 GHz). Both substantially exceed the 400 MHz minimum target.

6.2.3 Insertion Loss Discussion

The simulated transmission coefficients ($S_{21} = -3.15$ dB, $S_{31} = -3.04$ dB) represent an average excess insertion loss of ≈ 0.14 dB beyond the ideal 3.01 dB, corresponding to 3.2% of power dissipated in conductor, dielectric, and radiation losses. For FR-4 with $\tan\delta \approx 0.02$, this is consistent with expectations. A low-loss substrate (e.g., Rogers RO4003C with $\tan\delta \approx 0.0027$) would reduce the excess loss to below 0.05 dB.

7. Parametric Sensitivity Analysis

Each key parameter was swept individually while monitoring S_{11}, S_{21}, S_{31}, and S_{23}. Sweep ranges and step sizes are listed in Table 4.

Table 4. Parametric sweep definitions.

Parameter	Range	Step	Primary Effect
L_0 (quarter-wave length)	18–22 mm	0.5 mm	Centre frequency shift ($f \propto 1/L$)
W_1 (70.7 Ω width)	1.5–2.5 mm	0.2 mm	S_{11} matching depth
R_{iso} (isolation resistor)	50–200 Ω	25 Ω	S_{23} isolation level
ε_r (permittivity)	3.0–4.3	0.26	Centre frequency ($f \propto 1/\sqrt{\varepsilon_{\text{eff}}}$)
h (substrate height)	1.2–2.0 mm	0.2 mm	Bandwidth and matching depth

7.1 Quarter-Wave Length (L_0): 18–22 mm, 0.5 mm steps

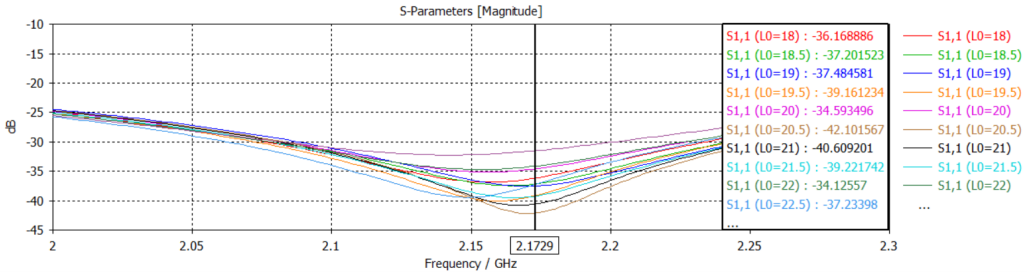


Figure 12. S_{11} vs. frequency for $L_0 = 18$–22 mm in 0.5 mm steps.

Increasing L_0 shifts the resonant frequency from ~ 2.175 GHz ($L_0 = 18$ mm) to ~ 2.152 GHz ($L_0 = 22$ mm), confirming the inverse relationship $f \propto 1/L$. The minimum S_{11} remains below -35 dB across the full sweep, demonstrating robust matching.

7.2 Transformer Width (W_1): 1.5–2.5 mm, 0.2 mm steps

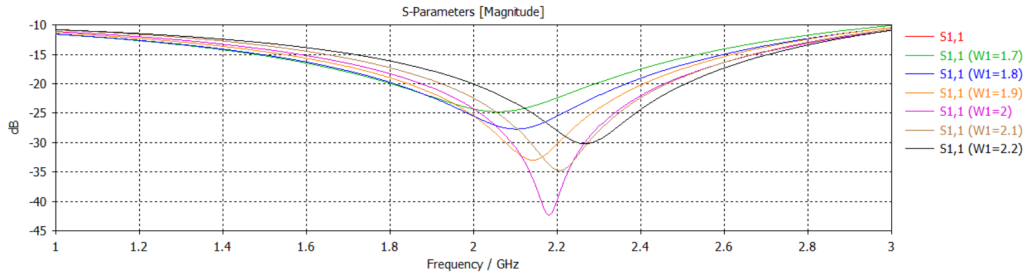


Figure 13. S_{11} vs. frequency for varying W_1.

Optimal matching occurs at $W_1 = 2.0\,\text{mm}$ with $S_{11} \approx -43\,\text{dB}$. Deviating by $\pm 0.5\,\text{mm}$ degrades matching by 10–15 dB, confirming that accurate control of the $70.7\,\Omega$ transformer width is critical.

7.3 Isolation Resistor (R_{iso}): 50–200 Ω

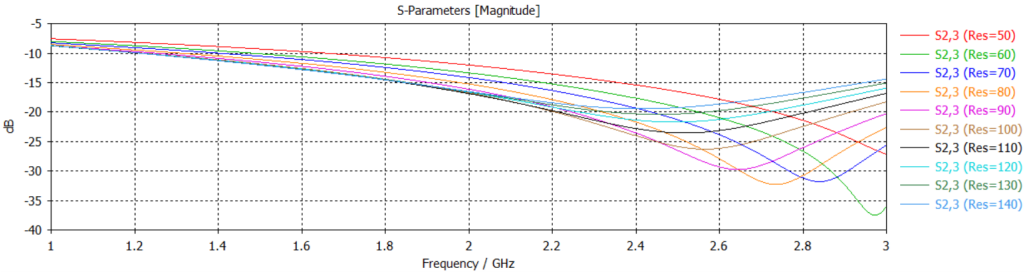


Figure 14. S_{23} vs. frequency for varying R_{iso}.

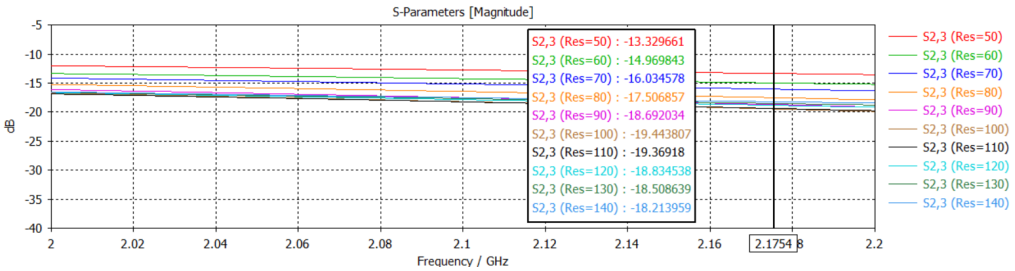


Figure 15. S_{23} detail — isolation resistor sweep.

At $R = 100\,\Omega = 2Z_0$, isolation reaches $S_{23} \approx -20\,\text{dB}$. Deviating to $50\,\Omega$ or $200\,\Omega$ degrades isolation by 5–10 dB, directly confirming the odd-mode prediction of Eq. (3.4).

7.4 Substrate Permittivity (ε_r): 3.0–4.3

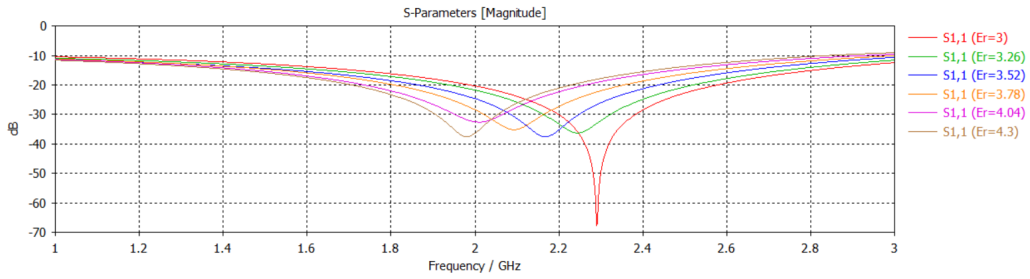


Figure 16. S_{11} vs. frequency for $\varepsilon_r = 3.0$–4.3.

Resonant frequency shifts from 2.288 GHz ($\varepsilon_r = 3.0$) to 1.98 GHz ($\varepsilon_r = 4.3$), yielding a sensitivity of -0.24 GHz per unit $\Delta\varepsilon_r$. This confirms $f \propto 1/\sqrt{\varepsilon_{\text{eff}}}$.

7.5 Substrate Height (h): 1.2–2.0 mm, 0.2 mm steps

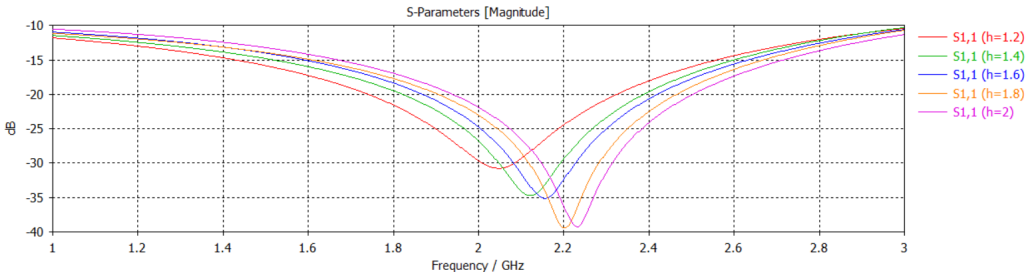


Figure 17. S_{11} vs. frequency for $h = 1.2$–2.0 mm in 0.2 mm steps.

Increasing h slightly shifts centre frequency upward (reduced ε_{eff}), deepens matching (best $S_{11} \approx -40$ dB at $h = 1.8$–2.0 mm), and slightly increases bandwidth.

8. Extension to Unequal Power Division

The Wilkinson topology generalises to unequal power splits. For a voltage ratio $K = V_3/V_2$ (power ratio $K^2 = P_3/P_2$, where Port 3 is the *high-power* port and Port 2 is the *low-power* port) [2]:

$$Z_2 = Z_0\sqrt{K(1+K^2)}, \qquad Z_3 = Z_0\sqrt{\frac{1+K^2}{K^3}} \tag{8.1}$$

$$R_{\text{iso}} = Z_0\left(K + \frac{1}{K}\right) \tag{8.2}$$

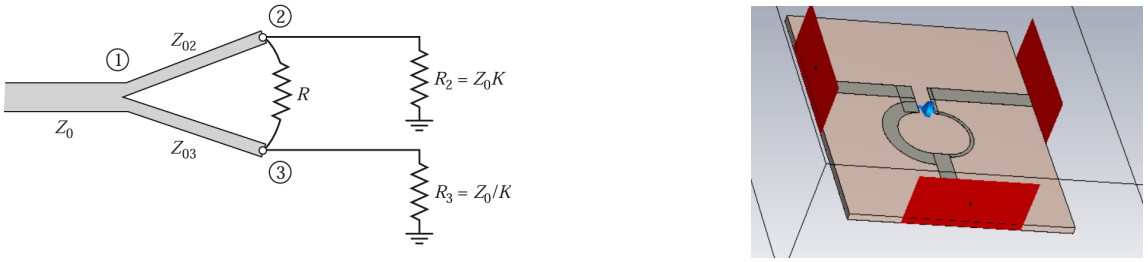
8.1 Design for 2:1 Power Split ($K = \sqrt{2}$)

For a 2:1 power ratio, $K = \sqrt{2}$. The theoretical transmission levels are:

- $S_{31} = 10\log_{10}(2/3) \approx -1.76$ dB (high-power port, Port 3)
- $S_{21} = 10\log_{10}(1/3) \approx -4.77$ dB (low-power port, Port 2)

Substrate change: A Rogers-type substrate ($\varepsilon_r = 3.55$, $h = 1.6\,\text{mm}$) was used for the unequal divider to demonstrate the flexibility of the Wilkinson topology across different substrate materials. This substrate has lower loss tangent than FR-4, which is preferable when the asymmetric impedance branches ($103\,\Omega$ and $51.5\,\Omega$) require tighter dimensional control. The design parameters are: $W_2 = 0.8\,\text{mm}$ ($\approx 103\,\Omega$), $W_3 = 3.4\,\text{mm}$ ($\approx 51.5\,\Omega$), $L_0 \approx 20\,\text{mm}$, and $R_{\text{iso}} \approx 106\,\Omega$ from Eq. (8.2).

8.2 Unequal Divider Layout and Results



(a) Unequal power divider schematic. **(b)** CST layout (top view).

Figure 18. Unequal power divider design.

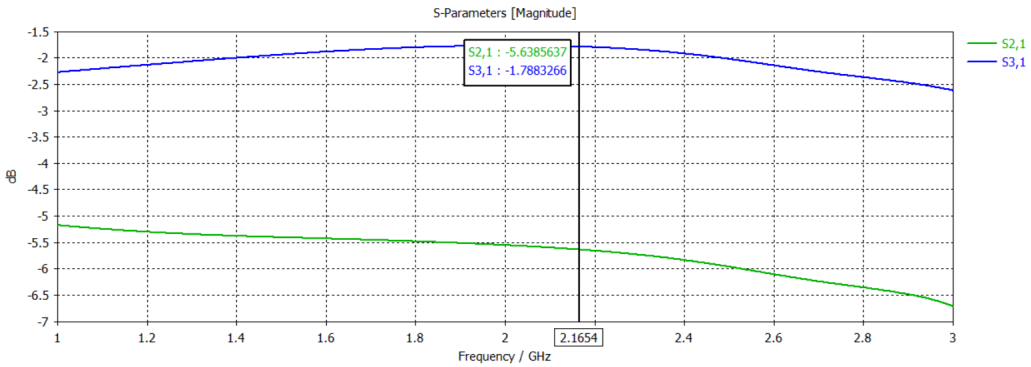


Figure 19. Unequal power division: S_{21} and S_{31} vs. frequency.

Results at 2.165 GHz**:** $S_{21} = -5.64\,\text{dB}$ (low-power port) and $S_{31} = -1.79\,\text{dB}$ (high-power port). The measured power ratio is $10^{(-1.79+5.64)/10} = 10^{0.385} \approx 2.43$, close to the target of 2.0. The slight deviation arises from different effective permittivities of the asymmetric microstrip lines.

9. Final Design Summary

9.1 Consolidated Geometry — Equal-Split Divider

Table 5. Final optimised dimensions for the equal-split Wilkinson divider.

Parameter	Symbol	Value
Centre frequency (design / simulated)	f_0	2.2 GHz / 2.15 GHz
System impedance	Z_0	50 Ω
Transformer impedance	Z_t	70.7 Ω
Isolation resistor	R_{iso}	100 Ω
Substrate	—	FR-4, $\varepsilon_r = 4.3$
Substrate height	h	1.6 mm
Copper thickness	T_s	17 μm
Quarter-wave arm length	L_0	$\approx$ 19 mm
50 Ω feed line width	W_0	$\approx$ 3.0 mm
70.7 Ω transformer width	W_1	$\approx$ 2.0 mm
Guided wavelength	λ_g	$\approx$ 76 mm
Free-space wavelength	λ_0	136.36 mm

9.2 Consolidated S-Parameter Results

Table 6. S-parameter results vs. acceptance criteria.

Parameter	Criterion	Ideal (2.2 GHz)	Practical (2.15 GHz)	Status
S_{21}	≈ -3 dB	-3.11 dB	-3.15 dB	✓ Pass
S_{31}	≈ -3 dB	-3.03 dB	-3.04 dB	✓ Pass
S_{11}	< -20 dB	-36.9 dB	-35 dB	✓ Pass
S_{23}	< -15 dB	-19.3 dB	-19.98 dB	✓ Pass
BW ($S_{11} < -20$ dB)	$>$ 400 MHz	—	$\approx$ 900 MHz	✓ Pass
BW ($S_{23} < -15$ dB)	$>$ 400 MHz	—	$\approx$ 1.3 GHz	✓ Pass

9.3 Unequal Divider Summary (2:1 Split)

Table 7. Unequal divider results summary.

Parameter	Theoretical	Simulated (2.165 GHz)
Substrate	$\varepsilon_r = 3.55$, $h = 1.6$ mm	Rogers-type
S_{21} (low-power)	-4.77 dB	-5.64 dB
S_{31} (high-power)	-1.76 dB	-1.79 dB
High-Z arm width (W_2)	—	0.8 mm ($\approx$ 103 Ω)
Low-Z arm width (W_3)	—	3.4 mm ($\approx$ 51.5 Ω)
Isolation resistor	106 Ω	106 Ω

10. Conclusion

This project designed, simulated, and validated both equal-split and unequal-split (2:1) two-way Wilkinson Power Dividers at 2.2 GHz using CST Studio Suite. The key findings are:

1. **Theoretical validation.** All simulated S-parameters agree closely with analytical predictions: S_{21} and S_{31} within 0.14 dB of the ideal -3.01 dB, $S_{11} = -35$ dB (req. < -20 dB), and $S_{23} = -19.98$ dB (req. < -15 dB).

2. **Frequency shift quantified.** The simulated optimum at 2.15 GHz (vs. design 2.2 GHz) represents a 2.3% downward shift, attributable to a $\sim$3.5% increase in effective electrical length from fringing fields.

3. **Three-port trade-off confirmed.** The deliberate sacrifice of losslessness via $R_{\text{iso}} = 2Z_0 = 100\,\Omega$ enables simultaneous matching and reciprocity; the resistor dissipates power only in odd mode.

4. **Parametric sensitivity characterised.** Sweep analysis quantified the impact of L_0, W_1, R_{iso}, ε_r, and h, providing engineering guidance for manufacturing tolerance and substrate selection.

5. **Unequal power division demonstrated.** The 2:1 variant on Rogers-type substrate ($\varepsilon_r = 3.55$) achieved $S_{31} = -1.79$ dB and $S_{21} = -5.64$ dB at 2.165 GHz.

6. **Broadband performance.** The 20 dB return-loss bandwidth of $\approx$ 900 MHz and 15 dB isolation bandwidth of $\approx$ 1.3 GHz make this design suitable for wideband applications including Wi-Fi, LTE, and radar front-ends.

References

[1] E. J. Wilkinson, "An N-way Hybrid Power Divider," *IRE Trans. Microwave Theory Tech.*, vol. 8, no. 1, pp. 116–118, Jan. 1960.

[2] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: John Wiley & Sons, 2012.

[3] R. E. Collin, *Foundations for Microwave Engineering*, 2nd ed. New York: McGraw-Hill, 1992.

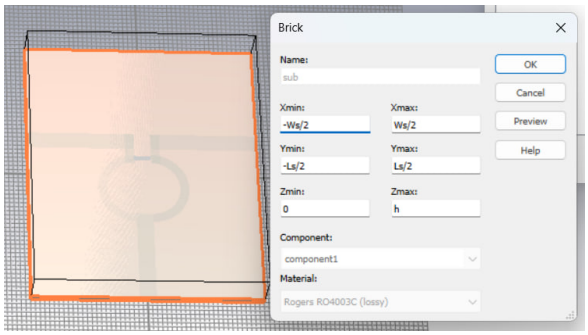
[4] G. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*. Norwood, MA: Artech House, 1980.

[5] S. B. Cohn, "A Class of Broadband Three-Port TEM-Mode Hybrids," *IEEE Trans. Microwave Theory Tech.*, vol. 16, no. 2, pp. 110–116, Feb. 1968.

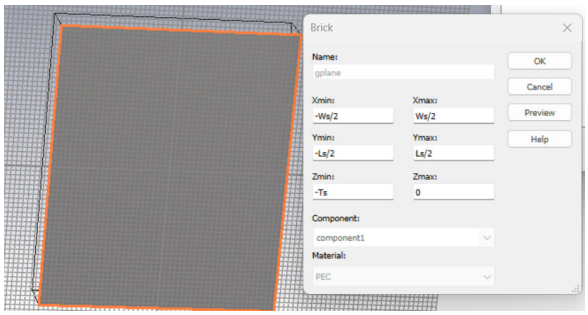
A. CST Design and Simulation Screenshots

The following figures document the step-by-step CST construction process, solver settings, and mesh configuration. They are provided for reproducibility.

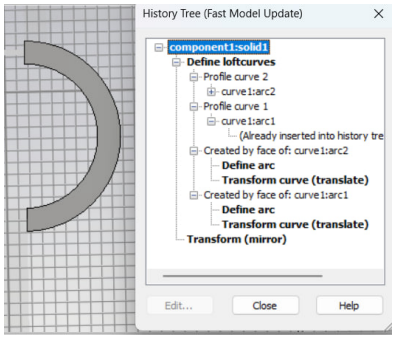
A.1 Substrate, Ground Plane, and Arch Construction



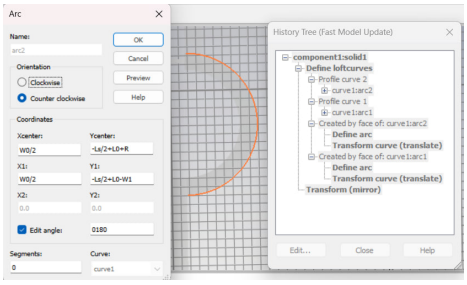
(a) Substrate geometry: FR-4 block with $\varepsilon_r = 4.3$ and height $h = 1.6\,\text{mm}$.



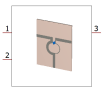
(b) Ground plane: PEC layer on the bottom face of the substrate.



(c) Arch design history tree showing Boolean operations for the curved $\lambda/4$ arms.



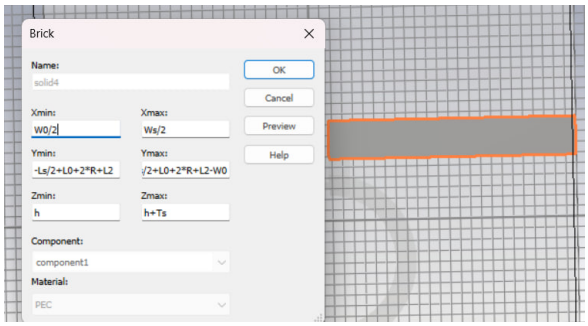
(d) Arch dimensional measurements: inner/outer radii and angular span.



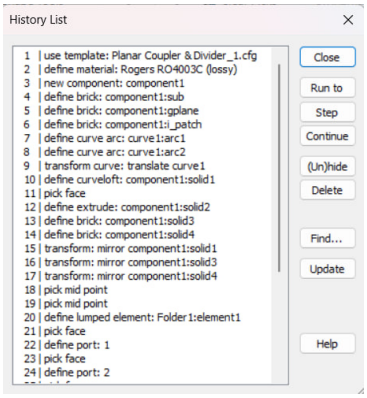
(e) Perspective view of the unequal divider variant.

Figure 20. Substrate, ground plane, and arch construction details.

A.2 Output Patches and Design History



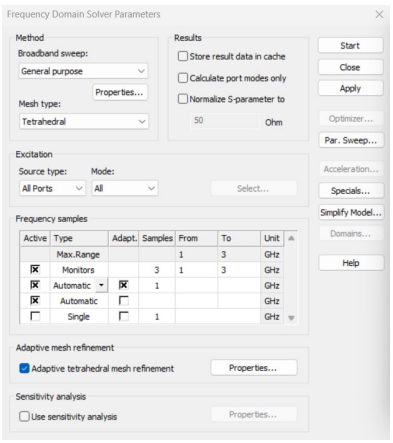
(a) Horizontal and vertical output patch geometry for Ports 2 and 3.



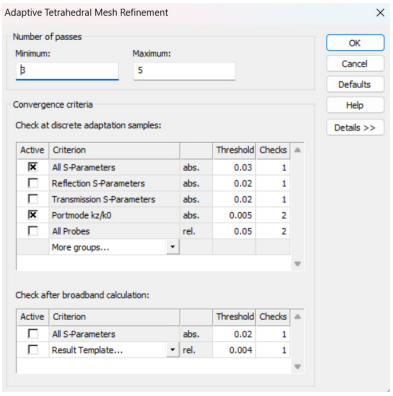
(b) Complete design history list in CST showing all modelling steps.

Figure 21. Output patch construction and design history.

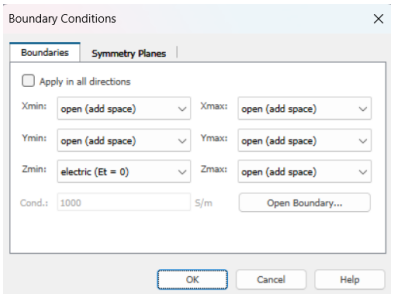
A.3 Solver Settings and Mesh Refinement



(a) Frequency-domain solver parameters: frequency range 1–3 GHz, tetrahedral mesh.



(b) Adaptive mesh refinement: convergence threshold $\Delta S < 0.03$, maximum 10 passes.



(c) Boundary conditions: Open (add space) on X, Y, $Z_{\max}$; Electric on $Z_{\min}$.

Figure 22. Solver configuration and boundary condition settings.